

DLD Lab 4

Sum $\rightarrow$ 4 minterms / maxterms
 Cout = 4 minterms / maxterms

q4-7.

minterms

Sum K-Maps

X \ Y C _{in}	0	1
00	0	1 $\leftarrow xy'cin'$
01	1 $\leftarrow x'y'cin$	0
11	1 $\leftarrow xy'cin$	0
10	1 $\leftarrow x'y'cin'$	0

$m(1, 2, 4, 7)$

$$\begin{aligned}\Sigma &= (xy'cin') + (x'y'cin) + (xy'cin) + (x'y'cin') \\ &= x'y'(cin + cin') + x'(y'cin + y'cin') \\ &= x'(y \oplus cin) + x(y \oplus cin)' = x \oplus y \oplus cin = Sum\end{aligned}$$

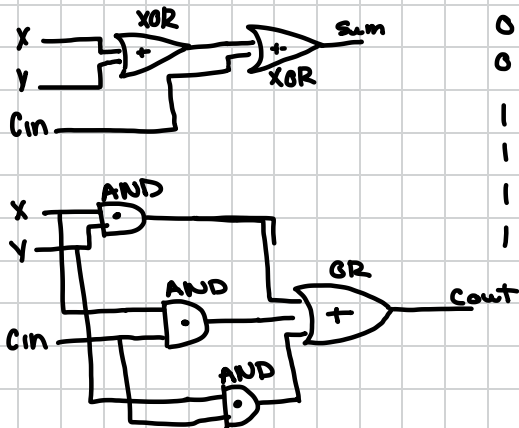
Cout K-Maps

X \ Y C _{in}	0	1
00	0	0
01	0	1 $\leftarrow cinx'y$
11	1 $\leftarrow cinx'y + cinxy$	1 $\leftarrow cinxy'$
10	1 $\leftarrow cinxy'$	0

$m(3, 5, 6, 7)$

$$\begin{aligned}\Sigma Cout &= cinx'y + cinx'y' + cinxy + cinxy' \\ &= y(cinx' + cinx) + cinx(y + y') \\ &= xy + xcin + ycin = Cout\end{aligned}$$

Schematics



X	Y	C _{in}	Cout
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Sum
maxterms

X \ Y C _{in}	0	1
00	0	0
01	0	0
11	0	0
10	0	0

$M(0, 3, 5, 6)$

$$\begin{aligned}& cinx'y' + cinx'y + cinxy + cinxy' \\ & (x+y+cin)(x+y'+cin')(x'+y'+cin)(x'+y+cin)\end{aligned}$$

Maxterm follow $(A+B)(A'+C)$
 Notice: White minterms for $AB+A'C$

Cout

X \ Y C _{in}	0	1
00	0	0
01	0	0
11	0	0
10	0	0

$M(0, 1, 2, 4)$

$$(cinx'y') + (cinx'y) + (cinxy') + (cinxy)$$

↓ it leads to

$$(x+y)(x+cin)(y+cin)$$

ALU- is a part of a computer's processor that performs the arithmetic and bitwise logic operations. This is where the full adder come into play as it is the core component to the arithmetic part of the ALU.